KUMANG SUBBA

Software Engineer

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BACKGROUND

Software engineer with professional experience building and maintaining production full-stack and backend applications using TypeScript, Node.js, NestJS, React, PostgreSQL, and cloud infrastructure. Strong foundation in C, C++, Go, Python, Linux, networking, and systems programming, with a particular interest in high-performance and low-level software.

Experienced across the software development lifecycle, including system design, implementation, testing, debugging, code review, CI/CD, deployment, and production troubleshooting. Comfortable working from high-level application development down to operating-system and networking concepts, and motivated by understanding how software behaves underneath abstractions.

PROFESSIONAL EXPERIENCE

Software Engineer
Vesko Oy | Jan 2025 – Jan 2026

- Developed and maintained production customer-facing web applications using React, TypeScript, and modern frontend development practices.
- Built and maintained backend services using Node.js and NestJS, including REST APIs and application integrations.
- Designed, implemented, tested, and deployed features across the full software development lifecycle.
- Worked with PostgreSQL through Azure Database for PostgreSQL.
- Designed and maintained CI/CD pipelines using GitHub Actions.
- Worked with Docker-based development and deployment environments and cloud infrastructure.
- Participated in debugging, testing, code reviews, peer programming, and technical discussions.
- Investigated application issues and worked systematically to identify root causes and implement fixes.
- Collaborated with engineers, designers, and stakeholders in an Agile environment.

Application Developer Intern
Kesko Oy | Jul 2024 – Dec 2024

- Developed an internal business application using React Native and Node.js.
- Implemented application features, backend integrations, and database functionality.
- Built and worked with backend APIs and application data flows.
- Collaborated with stakeholders to understand requirements and translate them into software features.
- Participated in testing, debugging, and code reviews.

TECHNICAL SKILLS

Programming Languages: C, C++, Go, Python, TypeScript, JavaScript, x86 Assembly
Systems Programming: Linux, C programming, memory management, pointers, processes, threads, concurrency, system calls, file descriptors, signals, CPU affinity, asynchronous I/O, performance analysis
Networking: TCP/IP, HTTP/HTTPS, TLS, sockets, non-blocking I/O, epoll, DNS, reverse proxies, load balancing, connection management
Backend Development: Node.js, NestJS, Express.js, Go, REST APIs, API integrations, authentication, validation, error handling
Frontend & Mobile: React, Next.js, React Native, Expo, Tanstack, React-router
Databases: PostgreSQL, MySQL, MongoDB, Redis, SQL, SQLite
Cloud & Infrastructure: Docker, Azure, AWS, Terraform, NGINX, GitHub Actions, CI/CD
Observability & Operations: Sentry, Prometheus, logging, metrics, monitoring, debugging, production troubleshooting
Development Tools: Git, Linux CLI, GDB, Make, ESLint, Prettier, Neovim
Engineering Practices: Testing, debugging, code reviews, Agile, Scrum, peer programming, continuous integration, continuous delivery

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PROJECTS & OPEN SOURCE

High-Performance Reverse Proxy — C / Linux
Built a reverse proxy from scratch in C to develop a deeper understanding of Linux networking and high-performance server architecture.

- Implemented connection handling using Linux epoll.
- Used non-blocking sockets for event-driven network I/O.
- Implemented HTTP request forwarding between clients and upstream servers.
- Worked directly with socket APIs, file descriptors, buffers, and connection state.
- Focused on efficient connection handling and memory management.
- Investigated networking and concurrency behaviour at the operating-system level.

NGINX
Actively study the NGINX source code to understand event-driven networking, connection management, configuration processing, and high-performance server architecture.

- Submitted an open-source pull request to NGINX proposing that the events{} configuration block be optional.
- Discussed the proposal with NGINX maintainers; the maintainers concluded that the change was unnecessary at the time.
- Continued studying the codebase as a way to understand production-grade C networking software

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EDUCATION

Bachelor of Business Administration in IT, Major in Developing Digital Services
Laurea University of Applied Sciences, 2023 – Present (189 / 210 credits, internship completed, Expected graduation: end of 2026)